

$P(\text{milk}, \text{cookies})$

$\rightarrow$ $P(\text{milk})$
 $P(\text{cookies})$

cookies

	<u>milk</u>			
	0	1	2	3
0	0	10	3	2
1	20	35	16	5
2	5	10	5	6
3	7	18	19	20

32

Σ

$$P(\text{cookies}=0) = 0 + 10 + 3 + 2 = \underline{15}$$

$$P(\text{cookies}=1) = \underline{70}$$

$$P(\text{cookies}=2) = \underline{26}$$

$$P(\text{cookies}=3) = \underline{64}$$

$P(\text{milk}, \text{cookies})$

milk

$P(\text{cookies})$

	0	1	2	3	
0	0	10	3	2	15
1	20	35	16	5	70
2	5	10	5	6	26
3	7	18	19	20	64
	32	...	...	...	

$P(\text{milk})$

cookies

$$P(\text{milk} | \text{cookies} = 1) = \frac{P(\text{milk}, \text{cookies} = 1)}{P(\text{cookies} = 1)}$$

$$P(\text{milk} = 3 | \text{cookies} = 1) = \frac{5}{70}$$

2 classes $\begin{cases} \text{apple} \\ \text{orange} \end{cases} \quad y$

Input x : average color

$$P(y | x) = \begin{matrix} P(\text{apple} | \text{color}) & \rightsquigarrow & P(\text{color} | \text{apple}) \\ P(\text{orange} | \text{color}) & \rightsquigarrow & P(\text{color} | \text{orange}) \end{matrix}$$

$$P(\text{apple} | \text{color}) = \frac{P(\text{color} | \text{apple}) P(\text{apple})}{P(\text{color})}$$

Statistical independence

		X			
		0	1	2	3
Y	0				
	1				
	2				
	3				

$$P(x, y) = 16$$

$$P(x) = 4$$

$$P(y) = 4$$

$$8$$

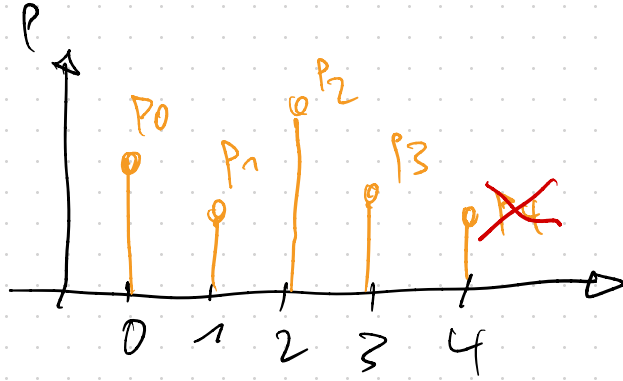
$$E_{x \sim p}[x] = \sum_x p(x) \cdot x$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \rightsquigarrow x_1 \sim P$$

$$\rightarrow x_2 \sim P$$

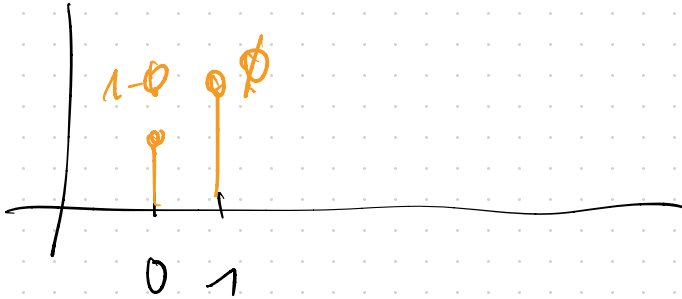
$$\rightarrow x_3 \sim P$$

Multi



$$p_4 = 1 - p_0 - p_1 - p_2 - p_3$$

$$p = \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix}$$



idx height [cm]

1	175
2	160
3	185
4	172

